DSC-002: Mathematical Methods

Lecture 2: **Prelims - Logic, Group theory**

Sugata Bag

Delhi School of Economics

August 3, 2026

1.1 Sentential Logic

Statements:

Assertions or *statements* have the property of being either *true* or *false*.
In sentential logic we consider only those sentences that are assertions.

Examples of Sentences:

(i) Sun rises in the east.

(ii) Two and two make five.

(iii) What time is it?

(iv) One ought not to tell lies!

◦ The first two are statements, but the last two are not.

◦◦ **Logical connectives** are operators used to combine one or more statements/propositions in propositional logic.
Five basic logical connectives: Negation, Conjunction, Disjunction, Conditional, and Bi-conditional.

1.2 Logical Connectives

(i) Negation

Negation ($\sim$): Negation of a true statement is false, and that of a false one is true. Let's see it in "Truth Table" below:

A	$\sim$**A**
T	F
F	T

Fact

- *The truth table like the one above can also be constructed for any "logical connectives" of the premises.*
- *If every entry in the final column is 'F' then the premises are inconsistent.*
- *If at least one entry is 'T' the premises are consistent.*

1.2 Logical Connectives

(ii) Conjunction

Conjunction ($\wedge$, also &): A conjunction of two statements is true *if and only if* both constituent statements are true.

A	**B**	**A** $\wedge$ **B**
T	T	T
T	F	F
F	T	F
F	F	F

Note that the word 'but' has about the same sense as 'and'.

○○ **Example:**
If **A**: "I am a good student."
And **B**: "I like my math teacher."
Then **A** $\wedge$ **B**: "I am a good student, and I like my math teacher."

1.2 Logical Connectives

(iii) Disjunction

Disjunction ($\vee$): In common usage 'or' is used both in exclusive and inclusive senses. In logic 'or' is always used in the inclusive sense.

A disjunction of two statements is false if and only if both of its constituents are false.

A	B	A $\vee$ B
T	T	T
T	F	T
F	T	T
F	F	F

Example: Let A: "It is raining." B: "It is cloudy."
The disjunction $A \vee B$ means: "It is raining or it is cloudy."

1.2 Logical Connectives

(iv) Conditional or Implication ($\rightarrow$)

Conditional or Implication ($\rightarrow$): A Conditional is a statement of the form $A \rightarrow B$ (read as: *If A then B*), where A and B are statements, further A is called the *antecedent* and B the *consequent*.

The conditional ($A \rightarrow B$) is true if A is false or both A and B are true. It is false only when A is true and B is false.

A	B	A $\rightarrow$ B
T	T	T
T	F	F
F	T	T
F	F	T

Example: Let $P : x > 0$ and $Q : x^2 > 0$.
If P is true, so is Q, but Q may be true even if P is not (e.g., $x = -2$).
Q: What about $Q \rightarrow P$? (this is "Converse" of original implication)

1.2 Logical Connectives

(v) Biconditional ($\leftrightarrow$)

Biconditional ($\leftrightarrow$): $A \leftrightarrow B$ (i.e. A *iff* B) is true iff A and B have the same truth-value.

A	B	A $\leftrightarrow$ B
T	T	T
T	F	F
F	T	F
F	F	T

Example 1 (Rule-Bound):
Let A: "You pass the pass/fail course." B: "You score 50% or higher."

$A \leftrightarrow B$: *"You pass the pass/fail course iff you score* 50% *or higher."*

Example 2 (Geometric Definition): Let
A: "A triangle has three equal sides." B: "A triangle has three equal angles."

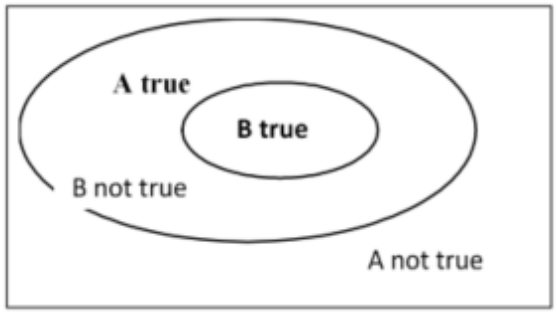
$A \leftrightarrow B$: *"A triangle has three equal sides if and only if it has three equal angles."*

◦ **Exercise:** Modify the previous example in last page to make it a biconditional. Hint - change the restriction on P!! Will it work??

1.3 - Necessity and Sufficiency

(i) Necessity: *Statement A is necessary for statement B* means:

- A must hold for B to hold
- B is true requires A to be true
- "A if B" or "A is implied by B" $(A \leftarrow B)$

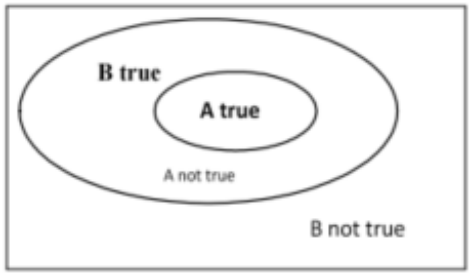


In the Venn diagram above: if A is not true, B must be not true (i.e. $\sim A \rightarrow \sim B$ (contrapositive of)).
(But that doesn't mean that if B is not true, A must be not true.)
Example: Every square is a rectangle $(B \rightarrow A)$, but not every rectangle is a square.

1.3 - Necessity and Sufficiency

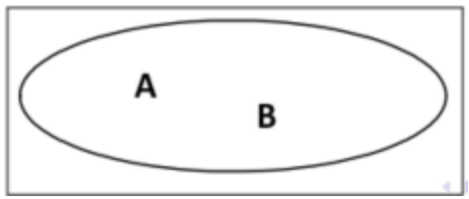
(ii) Sufficiency: *Statement A is sufficient for statement B* means:

- If A holds, B must hold
- "A only if B" $(A \rightarrow B)$



If B is not true $\Rightarrow$ A must be not true (i.e. $\sim B \rightarrow \sim A$ (Contrapositive)).
Example: Every multiple of 10 is even $(A \rightarrow B)$, but not every even number is a multiple of 10.

(iii) Both Necessity & Sufficiency: $A \leftrightarrow B$



1.4 - Logical Connectives

Proof Techniques for $A \to B$: A is true $\Rightarrow$ B must be true

- Here, A is called "*premise*" and B is called "*conclusion*"

- **Constructive / Direct proof:** Assume that A is true, deduce various consequences of that, and use them to show that B must hold.
- **Contrapositive proof:** Assume that B does not hold, then show that A cannot hold.

  $\sim B \to \sim A$ is the contrapositive of $A \to B$.

Think of some examples - numerical or real life cases.

1.4 - Logical Connectives

Definition (Claim)

A **statement** and its **Contra-positive** are equivalent.

Proof.

Suppose $P \rightarrow Q$ is true. Also, suppose Q is false.
Then P must be false, for if not, then via $P \rightarrow Q$, we would have Q be true, contradicting our assumption that Q is false.

On the other hand, suppose $P \rightarrow Q$ is false.
It follows that P is true and Q is false (for if P is false, then $P \rightarrow Q$ is vacuously true).
But then, $\sim Q \rightarrow \sim P$ cannot be true. □

Try to visualize the above in a truth table.

Contra-positive Truth table

P	Q	$\sim P$	$\sim Q$	$P \rightarrow Q$	$\sim Q \rightarrow \sim P$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

1.4 - Logical Connectives

Definition (Claim)

$Q \to P$ is called the **converse** of $P \to Q$.

Fact

There's no relationship between a statement and its converse.

Example

- True: If $x > 0$ then $x^2 > 0$
- Converse false: If $x^2 > 0$ then $x > 0$ (counterexample: $x = -1$)
- On the other hand, if some statement and its converse are both true, we say P if and only if Q or $P \leftrightarrow Q$.

1.5 - Tautology

i. **Tautology:** A tautology is a statement which is always true for all logically possible combinations of truth-values of the constituent *atomic* statements.

- Example: "It is raining or it is not raining."
  This sentence is always true, regardless of the weather.
- Logical example: $A \ \vee \sim A$ :: (A or not A – always true)
- Note: a sentence is *atomic* if it contains no sentential connectives.

ii. **Tautological Implication:**
'A' tautologically implies 'B' if and only if $(A \rightarrow B)$ is a tautology.

- Logical example: $A : x > 5, \quad B : x > 3$
  i.e. $A \rightarrow B$ means "If $x > 5$, then $x > 3$"
- Sentential example:
  "If a number is greater than 5, then it is greater than 3."

1.5 - Tautology, Inference

iii. **Tautological Equivalence:**
'A' is tautologically equivalent to 'B' *iff* $(A \leftrightarrow B)$ is a tautology.

- Logical example: $A : (x > 5) \wedge (x < 10), \quad B : 5 < x < 10$
- Sentential example: *"The number is greater than 5 and less than 10"* is equivalent to *"The number is between 5 and 10 (exclusive)."*

Examples of tautologies:

(i). $A \rightarrow A$

(ii). $A \rightarrow (A \vee B)$

(iii). $(A \wedge B) \leftrightarrow (B \wedge A)$

(iv). $(A \vee B) \leftrightarrow (B \vee A)$

(v). $(A \wedge (B \wedge C)) \leftrightarrow ((A \wedge B) \wedge C)$

◦ **Inference:** Q logically follows from P if $P \rightarrow Q$ is a tautology, or *iff* the conditional $P \rightarrow Q$ is universally valid.

◦ Nice reference: Lecture notes by Matthew Towers

1.6 - Quantifiers

Two quantifiers: *for all* $(\forall)$ and *there exists* $(\exists)$.

- Suppose, the statement P : *for all* $a \in A$, property $\Pi(a)$ holds.
  - The negation of P (i.e. $\sim P$) is the statement::
    - There exists $(\exists)$ *at least* one $a \in A$ s.t. property $\Pi(a)$ does not hold.

Example: Statement P: for every $x \in \mathbb{R}, x^2 > 0$.
Therefore, $\sim P$: there exists $x \in \mathbb{R}$ s.t. $x^2 \leq 0$.
Q: there exists $b \in B$ s.t. property $\Theta(b)$ holds.
Thus, $\sim Q$: for all $b \in B$, property $\Theta(b)$ does not hold.

Note. Order of quantifiers matters. e.g.
$\forall\ x > 0, \exists\ y > 0$ s.t. $y^2 = x$,
says that every positive real has a positive square root.
This is not the same as:
"$\exists\ y > 0$ s.t. $\forall\ x > 0,\ \ y^2 = x$ " ,
$\Longrightarrow$ some number y is the common square root of every positive real.

Modern Algebra

2.1 - Binary Composition/Operation

Definition (Binary Composition)

Given a set S, any *mapping* of $S \otimes S$ into S is called a bin. comp. on S.

I.e. An operation $\circ$ (or $*$) which when applied to two elements of a set S gives a unique element, also of the same set, is called a binary operation

- Thus given any set S, let $\circ$ be a *bin.comp.* on S.
  If $a, b \in S$ so that $(a, b) \in S \otimes S$ and $\exists$ a unique $c \in S$ such that $c = a \circ b = \circ(a, b) \in S$.
- This is **Closure Property** and the set S is *closed* w.r.t. operation $\circ$.

Algebraic Structure

Definition (Algebraic Structure:)

A non-empty set together with one or more than one operations is called *algebraic structure*. Ex: $(\mathbb{N}, +)$, $(\mathbb{Z}, +, -)$

Examples

- Addition is a binary operation on the set of even numbers, but is not on the set of odd natural numbers. Why?
- Subtraction is a binary operation the set of all integers, positive and negative, but not, if be limited to only positive integers.

Laws of Binary Operations

Associative law: A binary operation $*$ on the elements of the set S is said to be associative, if and only if, for any three elements $a, b, c \in S$, $a * (b * c) = (a * b) * c$.

Commutative law: A binary operation $\circ$ on the elements of the set S is commutative, if and only if, for any two $a, b \in S$, $a \circ b = b \circ a$

Distributive law : Two binary operations $\circ$ and $*$, say, on elements of the set S are said to be distributive, iff, for every element $a, b, c \in S$, .

. $a \circ (b * c) = (a \circ b) * (a \circ c)$ [left distributive]

. $(b * c) \circ a = (b \circ a) * (c \circ a)$ [right distributive]

Identity element. A set S is said to possess an identity element with respect the binary operation o, if there exists an element $I \in S$ with property that $I \circ x = x \circ I = x$, for every $x \in S$.

2.2 Group, Ring, Field

Definition (Group)

A **group** G is a nonempty set G along with **one** binary operation $\star$ s.t.
(i) for every pair $g, h \in G$, we have $g \star h \in G$ [:: **closure**]
(ii) for every 3 elements $g, h, j \in G$, we have
. $g \star (h \star j) = (g \star h) \star j$ [:: **associativity**]
(iii) There exists a unique element $e \in G$ s.t. for all $g \in G$,
. $g \star e = e \star g = g$. (Here, e is the **identity** element)
(iv) For every $g \in G$, there exists a unique element $h \in G$ s.t.
. $g \star h = h \star g = e$. (Here, h is the **inverse** of g, written g^{-1}).

If, $\forall g, h \in G$, $g \star h = h \star g$, then G is a commutative or **Abelian** group.

Examples

Real or complex numbers with addition is group.
But, multiplication of reals is not a group, since 0 does not have an inverse under multiplication.

2.2 Group, Ring, Field

Definition (Ring)

A **ring** G is a set with **at least** 2 elements, along with **2 operations**, addition $(+)$ and multiplication $(\cdot)$, such that -

(i) G with $+$ is a *commutative* group with identity element called 0 (zero), where 0 is the identity element under $+$.

(ii) G is closed and *associative* under multiplication $(\cdot)$.
There is a multiplicative *identity* element, called 1, s.t. $1 \neq 0$.

(iii) $\forall g, h, j \in G$, $g \cdot (h + j) = (g \cdot h) + (g \cdot j)$, and
. $(g + h) \cdot j = (g \cdot j) + (h \cdot j)$.

2.2 Group, Ring, Field

Definition (Field)

Field: Further, a field is a ring where the multiplication is commutative and every non-zero element has a multiplicative inverse

(i.e. if the non-zero elements of G form a commutative group under multiplication with 1 as identity element, then G is called a Field).

Example

Note: There are rings that are not fields.
$(\mathbb{R}; +, \cdot)$ and $(\mathbb{Q}; +, \cdot)$ are examples of fields,
$(\mathbb{Z}; +, \cdot)$ is an example of a Ring that's not a Field.

Vector Spaces

Definition

A **vector space** V over a field F of scalars, is a set with 2 operations -
- vector addition (called, an **internal composition**) and
- multiplication by scalar (called, an **external composition**) s.t. -

(i) V with $+$ is a commutative group with identity element $0 \in V$

(ii) If $v \in V$, $cv \in V, \forall c \in F$; and for $c_1, c_2 \in F$, $(c_1 c_2)v = c_1(c_2 v)$; $1 \in F$, the multiplication identity element for field F, satisfies $1v = v, \forall v \in V$.

(iii) For all $c \in F$ and $v_1, v_2 \in V$, $c(v_1 + v_2) = cv_1 + cv_2$, and for all $c_1, c_2 \in F$ and $v \in V$, $(c_1 + c_2)v = c_1 v + c_2 v$.

Note that we use the same symbol $+$ for both vector addition and scalar addition but they are different things.

Example

(1). $\mathbb{R}^n$ over the field $\mathbb{R}$, with vector addition defined as follows: $(x_1, ..., x_n)^T + (y_1, ..., y_n)^T = (x_1 + y_1, ..., x_n + y_n)^T$.
And scalar multiplication as $c(x_1, ..., x_n)^T = (cx_1, ..., cx_n)^T$.
Notice geometric interpretation (**Parallelogram Law** etc.).

$1 \in \mathbb{R}$ is the scalar-vector multiplication identity.

(2). $\mathbb{C}^n$ over $\mathbb{C}$.
Interpret c, x_i, y_i in (1) as complex numbers, and addition and multiplication as that of complex numbers.
Note that $1 + 0i$ is the scalar-vector multiplicative identity.

(3). $\mathbb{C}[a, b]$ over $\mathbb{R}$, where $[a, b] \subseteq \mathbb{R}$ and $\mathbb{C}[a, b]$ is the set of all real-valued, continuous functions with domain $[a, b]$.

It would be good if you check that the properties of a vector space are satisfied for the above examples.

- Vector spaces are also called linear spaces.
- Vectors are simply the elements of a linear space.

Logic Exercises

1. Truth Tables and Connectives

1. Construct truth tables for:
   - $(A \to B) \wedge (\sim B \to \sim A)$
   - $(A \vee B) \wedge \sim (A \wedge B)$ (Exclusive OR)
2. For statements:
   - P: "It is raining"
   - Q: "The ground is wet"

   Express in symbolic form:
   - "If it's not raining, then the ground isn't wet"
   - "The ground is wet if and only if it's raining"
3. Prove or disprove: $(A \to B)$ is equivalent to $(\sim A \vee B)$

Group Theory Exercises

2. Algebraic Structures

1. Which of these are groups? Justify.
   - $(\mathbb{Z}, +)$
   - $(\mathbb{R}, \times)$
   - $(\{1, -1, i, -i\}, \times)$ where $i^2 = -1$
2. For $G = \{f : \mathbb{R} \to \mathbb{R} | f(x) = ax + b, a \neq 0\}$ with function composition:
   - Show closure holds
   - Find the identity element
   - Find the inverse of $f(x) = 2x + 3$
3. Prove: In any group $(G, \star)$, the identity element is unique.

Vector Space Exercises

3. Vector Spaces and Fields

1. Verify whether these are vector spaces over $\mathbb{R}$:
   - The set of all 2×2 symmetric matrices
   - The set $\{(x, y, z) \in \mathbb{R}^3 | x + y + z = 1\}$
2. For the field $\mathbb{Z}_5$ (integers mod 5):
   - Construct addition and multiplication tables
   - Find multiplicative inverses of all non-zero elements
3. Let $V = C[0, 1]$ (continuous functions on [0,1]). For $f, g \in V$ and $c \in \mathbb{R}$, verify:
   - $(f + g)(x) = f(x) + g(x)$ is in V
   - $(cf)(x) = c(f(x))$ is in V

ANSWERS: Conditional Example

Example: Let $P : x > 0$ and $Q : x^2 > 0$ If P is true, so is Q, but Q may be true even if P is not (e.g., $x = -2$).

Truth Table for $P \to Q$:

x	$P : x > 0$	$Q : x^2 > 0$	$P \to Q$
2	T	T	T
-2	F	T	T
0	F	F	T

Q: What about $Q \to P$?

Truth Table for $Q \to P$:

x	$Q : x^2 > 0$	$P : x > 0$	$Q \to P$
2	T	T	T
-2	T	F	F
0	F	F	T

Answer: The conditional $Q \to P$ is **false** when $x = -2$, since Q is true but P is false. Otherwise, it is true.

Question: What about the biconditional $P \leftrightarrow Q$?

Truth Table:

x	$P : x > 0$	$Q : x^2 > 0$	$P \leftrightarrow Q$
2	T	T	T
-2	F	T	F
0	F	F	T

Answer: $P \leftrightarrow Q$ is true if both statements have the same truth value. It is false for $x = -2$.

Question: If we change P to $P : x \neq 0$, will $P \leftrightarrow Q$ always be true?

Truth Table:

x	$P : x \neq 0$	$Q : x^2 > 0$	$P \leftrightarrow Q$
2	T	T	T
-2	T	T	T
0	F	F	T

Answer: Yes. If P is redefined as $x \neq 0$, then P and Q are logically equivalent for all real x, and $P \leftrightarrow Q$ is always true.

Example: Necessity and Sufficiency

Statements:

- A: The person is a father.
- B: The person is a man.

Is being a man necessary for being a father?

- Yes. Every father is a man (**necessity**).
  $A \rightarrow B$ (Father $\rightarrow$ Man)
- Contrapositive: If not a man, then not a father.
  $\sim B \rightarrow \sim A$

Is being a man sufficient for being a father?

- No. Not every man is a father (**not sufficient**).

Both Necessity and Sufficiency?

- No. Not all men are fathers, so the statements are not equivalent.

Summary:

Being a man is **necessary** but **not sufficient** for being a father.